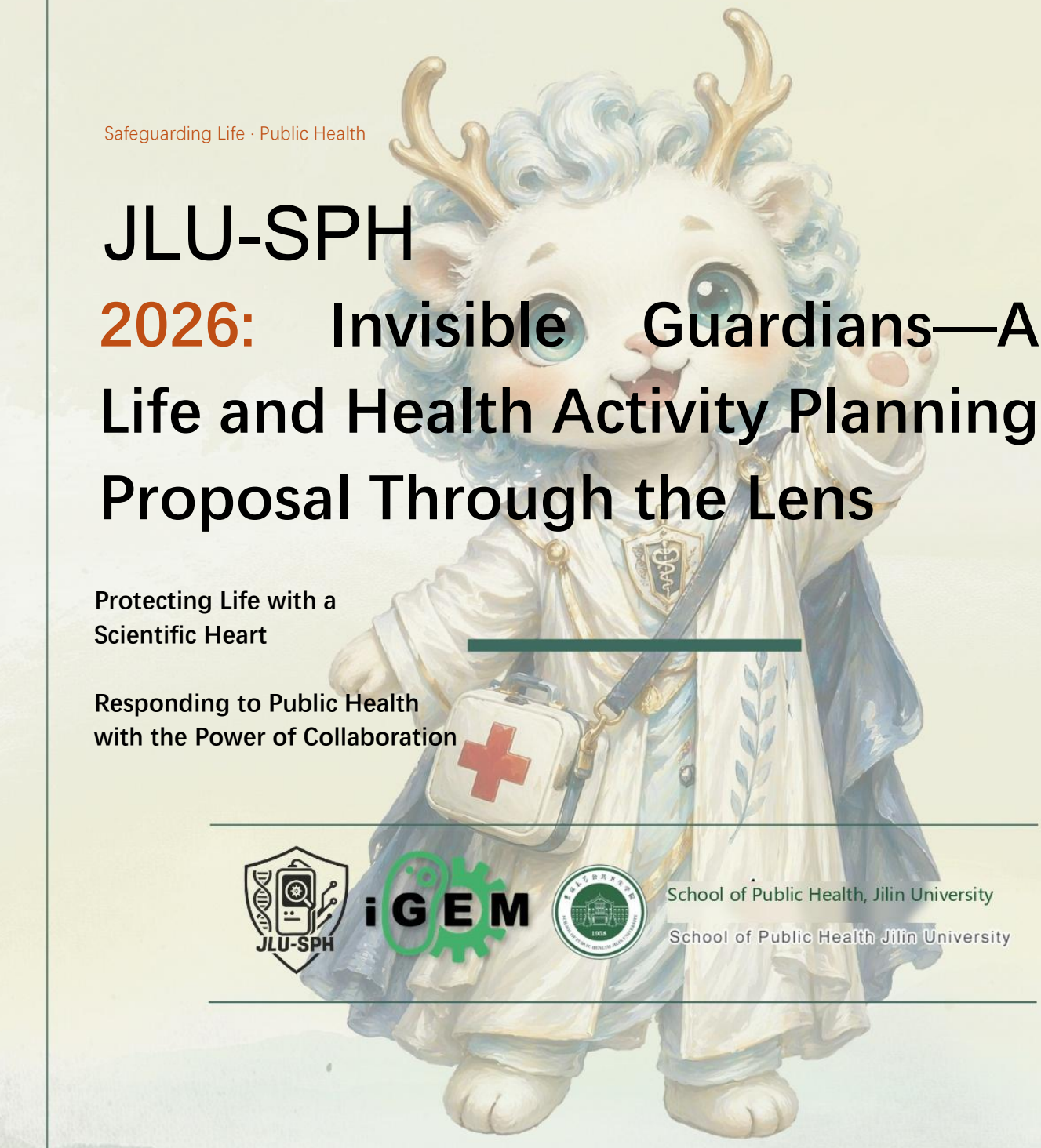
Safeguarding Life · Public Health

JLU-SPH 2026: Invisible Guardians—A Life and Health Activity Planning Proposal Through the Lens



Protecting Life with a Scientific Heart

Responding to Public Health with the Power of Collaboration

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I. Basic Information about the Event

- Event Title: Invisible Guardians: Life and Health through the Lens
- Event Theme: Using short film production and creation as a medium, this event connects natural life, school-based epidemic prevention, and synthetic biology-based detection, providing immersive respiratory health science education for secondary school students.
- Event Type: ☑ Science Education and Popularization ☐ Public Welfare Service ☐ Cultural Promotion ☐ Community Service ☐ Other:
- Organizer: JLU-SPH Team

II. Objectives of the Event

1. Core Objective

By leveraging visual media creation to lower the threshold of understanding regarding synthetic biology and respiratory infectious disease prevention knowledge, this activity aims to overcome the limitations of unidirectional classroom-based science communication. It seeks to guide secondary students toward establishing an integrated health awareness framework encompassing "natural life–daily protection–technological innovation detection," thereby improving the youth science popularization system of the project.

2. Specific Objectives

○ Knowledge/Cognition: Recognize the diversity of life, understand the transmission routes of respiratory infectious diseases, master conventional school-based protection measures, and comprehend the working principles and application value of synthetic biological sensors in pathogen detection.

○ Behavior/Skills: Develop skills in visual framing, filming, editing, scriptwriting, and team collaboration; learn to popularize life and health knowledge through short video formats.

○ Emotion/Attitude: Foster a sense of reverence for life and proactive protection awareness, reduce the perceived distance to frontier biotechnology, and encourage students to educate family members about health protection knowledge.

3. Anticipated Outcomes

○ Coverage: 60–100 secondary school students

○ Satisfaction Goal: Over 90%

○ Other Specific Indicators: Production of over 20 completed science popularization short films; collection of activity feedback through questionnaires; development of standardized filming storyboard templates and science communication script material packages; compilation of an outstanding short film collection for campus-wide exhibition and screening.

III. Target Audience

- Main Audience: Junior and senior high school students
- Age Range: 12–18 years old
- Participant Scale: ☑ Multi-class joint participation, totaling 60–100 participants / ☐ Other: / ☐ To be determined

- Participant Characteristics: High empathy, regular interest in filming and editing short videos; possess basic biological knowledge but find it challenging to connect daily epidemic prevention with synthetic biology-driven innovation; highly motivated to participate in creative experiential science popularization activities.

IV. Event Time and Venue

- Event Date: June 2, 2026
- Event Duration: ☑ 2 hours / ☐ 1 hour / ☐ Half day / ☐ Other:
- Specific Time Slot: After school in the afternoon, 14:30–16:30
- Event Venue: Multimedia classroom in the partner secondary school, school greenbelt areas, teaching building corridors

V. Content and Flow of the Event

[Event Process Overview]

Stage	Time	Session Name	Specific Content	Responsible Personnel	Required Materials
Stage	20 min	Venue Preparation and Material Distribution	Indoor equipment debugging, placement of science demonstration models, distribution of storyboard templates and filming props	Support Staff, Teaching Assistants	Computers, Projectors, Science Display Boards, Storyboard Scripts, Filming Props
Stage	10 min	Free Grouping by Participants	Students autonomously form teams and assign filming/on-camera/scriptwriting/editing roles	Moderator, Teaching Assistants	Grouping Cards, Markers
Main Stage	20 min	Introduction to Science Topics and Example Demonstrations	Lecturer explains respiratory health and the principles of synthetic biological detection; demonstration short films are played	Lead Lecturer, Moderator	Popular Science PPT, reference video clips, biological models
Main Stage	40 min	Group Outdoor Filming	Each group films scenes on campus related to nature, epidemic prevention, and scientific innovation	All Teaching Assistants	Mobile phone holder, supplementary lighting, reference shooting images

Main Stage	30 min	Short video editing and creation of popular science narration	Group integration of materials, pairing with popular science scripts to complete video editing	Teaching assistant, technical guidance	Popular science script materials, tablet/computer
Final stage	20 min	Offline work sharing and peer review	Each group presents their short video on stage; teacher-student interaction and feedback; lecturer summarizes and enhances understanding	Overall coordinator, host	Audio system, projection devices, scoring sheets
Wrap-up stage	10 min	Questionnaire collection, venue restoration	Collect questionnaires, gather short video materials, clean the venue, store materials	Support Staff, Teaching Assistants	Paper questionnaires, storage boxes, cleaning tools

【Detailed Description of Each Segment】

Segment 1: Introduction to popular science and case demonstration (20 minutes)

- Details: The lecturer introduces the hazards of respiratory infectious diseases, and the synthetic biology detection device in this project to provide scientific popularization, explaining the three-layer protection logic; plays a demonstration short video produced by the team, explains narrative ideas through the lens, and distributes storyboard templates.
- Interaction mode: ☑ Explanation ☑ Demonstration ☑ Q&A
- Key phrases: The lens is our tool for expressing science; every grass and tree is life; masks and hand washing are protection; synthetic biology is the invisible defensive line.
- Notes: Control the density of technical terms, use more real-life examples, reserve time for answering questions.

Segment 2: Grouped outdoor shot selection and filming (40 minutes)

- Details: Students are grouped for filming: recording plant life to capture nature; documenting hand washing, mask wearing, and ventilation in campus scenes to illustrate daily protection; filming bacteriophage , and biosensor models to explain technological innovation in detection; teaching assistant provides full guidance in shot selection and composition.
- Interaction mode: ☑ Hands-on ☑ Group collaboration ☐ Explanation ☐ Games
- Key phrases: Capture authentic campus scenes, use the lens to record the invisible guardians of health around us.
- Notes: Define safe shooting areas, avoid students chasing or roughhousing, remind to safeguard shooting equipment.

Segment 3: Short video editing and popular science narration production (30 minutes)

- Details: Each group integrates filming materials, selects supporting popular science scripts to adapt narration, completes editing, constructing a complete narrative on "Cherish life—implement daily protection—technology innovation guarding health." Technical teaching assistants help solve editing operation issues.

Interaction mode: ☑ Hands-on ☑ Group discussion

- Key phrases: Present your understanding of health protection through short videos to more classmates.
- Notes: Provide simplified editing operation guidelines in advance to ensure each group can complete their video.
- Segment 4: Offline work sharing and peer review (20 minutes)

Details: Each group displays their short video via projection in turn; group members share creative ideas; teachers and students provide on-site feedback and scores based on visual creativity, completeness of popular science, and narrative logic; lecturer summarizes activity, linking the three-layer protection system of nature, daily life, and scientific innovation.

- Interaction mode: ☑ Sharing ☑ Q&A ☑ Peer review

Key phrases: Every short video is a unique response from youth, safeguarding respiratory health from their perspective.

- Notes: Feedback focuses on encouragement, improvement suggestions are given for deficiencies.
- Segment 5: Questionnaire collection, venue restoration
- Details: Distribute feedback questionnaires on site, collect all student-produced short videos; dismantle display materials, clean indoor and outdoor venues, and store popular science models and shooting props.

VI. Personnel Arrangement and Division of Duties

Overall coordinator: 1 person; responsible for coordination with the school, overall process control, handling emergencies

Host: 1 person; process integration, atmosphere facilitation, hosting work sharing segment

Main speaker for popular science: 1 person; responsible for scientific knowledge explanation, interpretation of demonstration videos, activity summary

Technical teaching assistant: 2 persons; guidance on editing, handling equipment problems, projection adjustment

On-site teaching assistant: 3 persons; group guidance for filming, maintaining order, distributing materials

Photography recorder: 1 person; full-event photographing highlights, archiving students' short videos

Logistics and support: 2 persons; venue setup, material storage, emergency response, cleaning and wrap-up

VII. List of Materials

【Promotion-related】

- Event Theme Poster: A3 color, 4 copies, to be posted at classrooms and key locations on campus
- Popular Science Leaflet: A5, 100 copies, to be distributed before student activities, providing basic information on epidemic prevention and a brief introduction to synthetic biology

【Teaching/Activity Category】

- Standardized Shooting Storyboard Template: Reference for Student Creation
- Compilation of Popular Science Scripts: Adapted to Middle School Students' Language Style
- Synthetic Biology Popular Science Model: On-site Filming and Live Explanation
- Short Film Peer Evaluation Scoring Sheet: For Scoring During Work Sharing
- Activity Feedback Questionnaire: Collecting Experience Feedback After the Event
- Mobile Portable Stand: For Students to Borrow for Stabilized Filming

[Logistics]

- Attendance Sheet
- Signing Pen: For Signing In, Filling Out Questionnaires, and Scoring
- First Aid Kit: Standard Configuration, 1 unit; For Emergency Response to Student Bumps and Abrasions
- Storage Box: For Storing Models and Paper Materials
- Cleaning Wet Wipes and Garbage Bags: Several Units, For Venue Cleaning After the Event

[Equipment]

- Laptop computer: 1 unit, for presenting PPT (PowerPoint presentations), demonstrating video clips, and screen-casting works
- Projector and Screen: 1 set; For Indoor Science Popularization and Short Film Display. Confirm with School Administration in advance.
- Sound System and Wireless Microphone: 1 set; For Amplified Presentation and Sharing Sessions
- Multi-Outlet Power Strip: 3 units; For Equipment Power Supply Extension

VIII. Risk Response Plan

- Personnel Risk: If students fall or feel unwell during outdoor filming, promptly use the First Aid Kit for initial treatment, contact the school nurse, and inform parents if the situation is severe.
- Venue Risk: In case of sudden heavy rain preventing outdoor filming, switch to indoor corridors or green corners for shooting, adjust the shooting sequence accordingly.
- Equipment Risk: If projection or sound equipment fails, or student filming devices are damaged, use tablets to play short films in rotation and substitute with backup stands/materials.
- Order Risk: Should filming groups disperse or there is excessive noise, teaching assistants will supervise at designated zones and allocate fixed filming time slots.
- Material Risk: If storyboards or science scripts are insufficient, print extra copies in advance and synchronize digital files online.

IX. Effect Evaluation

- Process Record: Photographic materials are the core basis for evaluation.
- Outcome Statistics: Collect the number of student works and activity feedback questionnaires.